
Test Case ID	Input Scenario	Expected Category	Expected Priority	Result
TC001	what tables exist in the database?	list_tables	P1	Pass
TC002	show schema of Employees table	get_schema	P1	Pass
TC003	show all employees	select_query	P2	Pass
TC004	show employees with salary greater than 55000	Filtered query	P1	Pass
TC005	show all records from NonExistentTable	Invalid Access	P3	Fail

Functional Testing:

MCP server processing (MCP Tool Calls):

- Verify `list_tables` tool correctly returns available tables (Employees, Departments, Projects).
- Verify `get_schema` tool returns schema details for valid tables (e.g., Employees).
- Verify `run_select` tool executes valid queries and returns expected rows.
- Verify invalid queries (e.g., `NonExistentTable`) are skipped and return structured error messages.
- Verify filtered queries (`salary > 55000`) produce correct subset of results.

Database Storage:

- Verify `chatdata.db` is created when MCP Server runs.
- Verify schema and query results are inserted into appropriate tables (e.g., Employees tables).
- Verify invalid queries do not corrupt database storage.
- Verify joins and aggregates are stored correctly when executed.

Output:

- Verify `output/results.csv` is created
 - Code Reference: Likely handled in `Tools` folder (CSV export utilities).
- Verify all processed tickets are stored
 - Code Reference: `Models/SelectQueryResult.cs` (query results model), `SqliteService.cs` (data persistence).

Workflow:

- Verify Thought step is generated
 - Code Reference: `ClaudeCompletionRequest.cs` (captures user query).
- Verify Action step is generated
 - Code Reference: `Tools` folder (MCP tool implementations: `list_tables`, `get_schema`, `run_select`).
- Verify Observation step is generated
 - Code Reference: `Models/SelectQueryResult.cs` (query results returned).
- Verify Final Answer is generated
 - Code Reference: `ClaudeCompletionResponse.cs` (final structured response back to Claude Desktop).

Dashboard Testing:

1. Verify dashboard loads correctly without errors.
2. Verify query results (tables, schema, employees) are displayed accurately.
3. Verify filters (e.g., salary > 55000) show correct subsets.
4. Verify responsiveness and smooth navigation across sections.

Error Handling:

1. Verify invalid table access returns clear error messages.
2. Verify invalid SQL syntax is handled gracefully.
3. Verify database connection failures are reported without crashing.
4. Verify all errors are logged for debugging.

Performance Testing:

1. Verify query execution time remains within acceptable limits.
2. Verify dashboard refresh is fast even with large datasets.
3. Verify concurrent queries from multiple users do not degrade performance.
4. Verify stress testing with thousands of records does not crash the MCP Server.